Theorem 5. For sparse attention:

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} 0 \qquad \epsilon^{\Delta x(k)} \xrightarrow[\epsilon_t^k \to 0]{\text{perturb k}} 0 \qquad \epsilon^{\Delta x(m)} \xrightarrow[\epsilon_t^m \to 0]{\text{perturb m}} \epsilon_{t^*}^m \tag{50}$$

i.e. the message is stable with respect to small interference in the queries and keys. Interference in the selected value is linearly transferred onto the message.

Proof. For sparse attention we have $a_t = \delta_{tt^*}$ for some t^*. For perturbations of q, the RHS of Eq. 39 becomes

$$\begin{aligned} \mathop{\mathbb{E}}_{a_t}\left[m_t \tilde{k}_t^T\right]\epsilon^q &= \sum_t a_t m_t \tilde{k}_t^T \epsilon^q \\ &= \sum_t \delta_{tt^*} m_t \tilde{k}_t^T \epsilon^q \\ &= m_{t^*} \tilde{k}_{t^*}^T \epsilon^q \\ &= 0 \end{aligned} \tag{51}$$

where the final step is because $\tilde{k}_{t^*} = k_{t^*} - \mathbb{E}_{a_t}[k_t] = k_{t^*} - \sum_t \delta_{tt^*} k_t = k_{t^*} - k_{t^*} = 0$. For perturbations of k_t, the RHS of Eq. 40 evaluates to 0 because

$$\begin{aligned} \mathop{\mathbb{E}}_{a_t}\left[\tilde{m}_t {\epsilon_t^k}^T\right] q &= \sum_t a_t \tilde{m}_t q^T \epsilon_t^k \\ &= \sum_t \delta_{tt^*} \tilde{m}_t q^T \epsilon_t^k \\ &= \tilde{m}_{t^*} q^T \epsilon_{t^*}^k \\ &= 0 \end{aligned} \tag{52}$$

where the final step is because $\tilde{m}_{t^*} = m_{t^*} - \sum_t \delta_{tt^*} m_t = m_{t^*} - m_{t^*} = 0$. For perturbations of m_t, the RHS of Eq. 41 evaluates to

$$\mathop{\mathbb{E}}_{a_t}\left[\epsilon_t^m\right] = \sum_t a_t \epsilon_t^m = \sum_t \delta_{tt^*} \epsilon_t^m = \epsilon_{t^*}^m \tag{53}$$

Theorem 6. For isotropic attention:

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} \langle m_t \tilde{k}_t^T \rangle_t \epsilon^q \qquad \epsilon^{\Delta x(k)} \xrightarrow[\epsilon_t^k \to 0]{\text{perturb k}} \langle \tilde{m}_t {\epsilon_t^k}^T \rangle_t \, q \qquad \epsilon^{\Delta x(m)} \xrightarrow[\epsilon_t^m \to 0]{\text{perturb m}} \langle \epsilon_t^m \rangle_t \tag{54}$$

N.B. isotropy requires $k_t = const$ or $q = 0$. Lemma 1: the update is stable to noisy q when $k_t = const$, or when $m_t \perp k_t$ (c.f. keys and messages from independent subspaces). Lemma 2: the update is stable to noisy k_t when $q = 0$, or when $m_t \perp \epsilon_t^k$. Lemma 3: the update is stable to noisy m_t when $\langle \epsilon_t^m \rangle_t = 0$. Other cases propagate linearly.

Proof. For isotropic attention we have $a_t = \frac{1}{T}$. For perturbations of q, the RHS of Eq. 39 is

$$\begin{aligned} \mathop{\mathbb{E}}_{a_t}\left[m_t \tilde{k}_t^T\right]\epsilon^q &= \sum_t a_t m_t \tilde{k}_t^T \epsilon^q \\ &= \frac{1}{T}\sum_{t=1}^{T} m_t \tilde{k}_t^T \epsilon^q \\ &= \langle m_t \tilde{k}_t^T \rangle_t \epsilon^q \end{aligned} \tag{55}$$

For lemma 1, we note that $k_t = const$ implies $\tilde{k}_t = 0$, and if $m_t \perp k_t$ then $\langle m_t \tilde{k}_t^T \rangle_t = \langle m_t k_t \rangle_t - \langle m_t \rangle_t \langle k_t \rangle_t = Cov(m_t, k_t) = 0$.

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For perturbations of k_t, the RHS of Eq. 40 is

$$\begin{aligned} \mathop{\mathbb{E}}_{a_t}\left[\tilde{m}_t {\epsilon_t^k}^T\right] q &= \frac{1}{T}\sum_{t=1}^{T} \tilde{m}_t {\epsilon_t^k}^T q \\ &= \langle \tilde{m}_t {\epsilon_t^k}^T \rangle_t q \end{aligned} \tag{56}$$

For lemma 2, this expression evaluates to 0 if $q = 0$, and if $m_t \perp \epsilon_t^k$ then $\langle \tilde{m}_t {\epsilon_t^k}^T \rangle_t = \langle m_t {\epsilon_t^k}^T \rangle_t - \langle m_t \rangle_t \langle {\epsilon_t^k}^T \rangle_t = Cov(m_t, {\epsilon_t^k}^T) = 0$.

For perturbations of m_t, the RHS of Eq. 41 evaluates to

$$\mathop{\mathbb{E}}_{a_t}\left[\epsilon_t^m\right] = \frac{1}{T}\sum_{t=1}^{T} \epsilon_t^m = \langle \epsilon_t^m \rangle_t \tag{57}$$

Theorem 7. Sensitivity of sparse attention to multiplicative perturbations $\epsilon^q = \kappa^q q$ and $\epsilon^k = \kappa_t^k k_t$ with $\kappa^q, \kappa_t^k \ll 1$. Circuit collapse occurs when $\exists\, t \neq t^*$ for which:

$$\frac{w_{t^*}}{w_t} \begin{cases} < \lambda_w & \text{if } w_t\left(1 + \kappa^q + \kappa_{t^*}^k\right) > 0 \\ > \lambda_w & \text{otherwise} \end{cases} \qquad \lambda_w \triangleq \frac{1 + \kappa^q + \kappa_t^k}{1 + \kappa^q + \kappa_{t^*}^k} \tag{58}$$

where temperature cancels in the fraction. **Attention is fully stable above the critical transition point** λ_w (c.f. $w_t\left(1 + \kappa^q + \kappa_{t^*}^k\right) > 0$). We see that query perturbations alone are insufficient, as they result in $\lambda_w = 1$. Lemma: consider the special case when all keys have similar length $k_t \approx const$, the attended token has $\theta_{t^*} \approx 0$, the keys are far-from-orthogonal s.t. $\theta_t \ll 1$, and $\kappa^q \approx 0$. Using $w_t \triangleq |q||k_t| \cos\theta_t$, circuit collapse occurs when $\exists\, t \neq t^*$ for which:

$$\frac{1}{2}\theta_t^2 \lesssim \kappa_t^k - \kappa_{t^*}^k \qquad \text{if } w_t\left(1 + \kappa_{t^*}^k\right) > 0 \text{ , otherwise reverse} \tag{59}$$

i.e. stability requires either well-separated keys s.t. $\theta_t \gg 0$, or small perturbations $\kappa_t - \kappa_t^* \ll 1$.

Proof. Apply $q \to q + \epsilon^q$ and $k_t \to k_t + \epsilon_t^k$ to $w_t = q^T k_t$, then we have $w_t \to w_t + \epsilon_w$ such that $\epsilon_t^w = q^T \epsilon_t^k + \epsilon^{q^T} k_t + \epsilon^{q^T} \epsilon_t^k$. For multiplicative perturbations we have $\epsilon^q = \kappa^q q$ and $\epsilon^k = \kappa_t^k k_t$, and so $\epsilon_t^w = \kappa_t^k q^T k_t + \kappa^q q^T k_t + \kappa_t^k \kappa^q q^T k_t$. Each term recovers a factor of $w_t = q^T k_t$, which we factor out to give $\epsilon_t^w = \left(\kappa^q + \kappa_t^k + \kappa_t^k \kappa^q\right) w_t$. The final term is subleading in the limit of small perturbations, and so

$$\epsilon_t^w \xrightarrow[\kappa^q, \kappa_t^k \to 0]{} \left(\kappa^q + \kappa_t^k\right) w_t + \mathcal{O}\left(\kappa^q \kappa_t^k\right) \tag{60}$$

Circuit collapse occurs when $w_{t^*} - w_t < \epsilon_t^w - \epsilon_{t^*}^w$ for some t. Substituting our limit for ϵ_t^w gives

$$w_{t^*} - w_t < \left(\kappa^q + \kappa_t^k\right) w_t - \left(\kappa^q + \kappa_{t^*}^k\right) w_{t^*} \tag{61}$$

and collecting terms gives

$$\left(1 + \kappa^q + \kappa_{t^*}^k\right) w_{t^*} < \left(1 + \kappa^q + \kappa_t^k\right) w_t \tag{62}$$

We then divide each side by $w_t(1 + \kappa^q + \kappa_{t^*}^k)$, taking care to reverse the sign of the inequality when this factor is negative, to give

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$$\frac{w_{t^*}}{w_t} \begin{cases} < \lambda_w & \text{if } w_t\left(1+\kappa^q+\kappa^k_{t^*}\right) > 0 \\ > \lambda_w & \text{otherwise} \end{cases} \qquad \lambda_w \triangleq \frac{1+\kappa^q+\kappa^k_t}{1+\kappa^q+\kappa^k_{t^*}} \tag{63}$$

which is the first expression in the theorem. We note that any temperature parameter cancels in the fraction, which means that the attention head cannot become more stable by reducing its temperature to become more sparse. λ_w has the limits

$$\lambda_w \xrightarrow[\kappa^q \to 0]{\text{keys only}} \frac{1+\kappa^k_t}{1+\kappa^k_{t^*}} \qquad \lambda_w \xrightarrow[\kappa^k_t, \kappa^k_{t^*} \to 0]{\text{query only}} \frac{1+\kappa_q}{1+\kappa_q} = 1 \tag{64}$$

meaning that query perturbations alone are insufficient, contributing only when they co-occur with perturbations on the keys. Write $w_t = |q||k_t|\cos\theta_t$ with $\theta_t = q \wedge k_t$, and the approximation of identical key norms $k_{t^*} = k_t \equiv k$ turns this into $w_t = |q||k|\cos\theta_t$. Then

$$\frac{w_{t^*}}{w_t} = \frac{|q||k|\cos\theta_{t^*}}{|q||k|\cos\theta_t} = \frac{\cos\theta_{t^*}}{\cos\theta_t} \tag{65}$$

Then $\theta_{t^*} = 0$ means that $\cos\theta_{t^*} = \cos 0 = 1$, and so $\frac{\cos\theta_{t^*}}{\cos\theta_t} = \frac{1}{\cos\theta_t} = \sec\theta_t$. We perform a Taylor expansion in θ_t to obtain

$$\frac{w_{t^*}}{w_t} \approx \sec\theta_t \approx 1 + \frac{1}{2}\theta_t^2 + \mathcal{O}\left(\theta_t^4\right) \tag{66}$$

which is valid when $\theta_t \ll 1$. This is true for any $t \neq t^*$ for which k_t is far from orthogonal with k_{t^*}. Substituting this into our circuit collapse condition, we have

$$1 + \frac{1}{2}\theta_t^2 < \frac{1+\kappa^k_t}{1+\kappa^k_{t^*}} \qquad \text{if } w_t\left(1+\kappa^k_{t^*}\right) > 0 \tag{67}$$

where we consider the case of $\kappa_q \approx 0$ for readability. Re-arranging gives

$$\frac{1}{2}\theta_t^2 \lesssim \kappa^k_t - \kappa^k_{t^*} \qquad \text{Circuit collapse when } k_t \text{ similar} \tag{68}$$

if $w_t(1+\kappa^k_{t^*}) > 0$, and we reverse the inequality otherwise. We have approximated the denominator on the RHS as $1+\kappa^k_{t^*} \approx 1$ for $\kappa^k_{t^*} \to 0$.

When $\theta_t \ll 1$, the LHS of Eq. 68 is small. This means that the attention head can tolerate only very small perturbations $\{\kappa^k_t, \kappa^k_{t^*}\}$. Therefore semantic subspaces must either have a highly orthogonal substructure s.t. $\theta_t \gtrsim 1 \; \forall \; t \neq t^*$, or be orthogonal s.t. $\kappa_t \ll 1 \; \forall \; t$.